

Part II — Regime Viability on the Northern Cod Stock: An Exact Certification of Obstruction Structure on Public Assessment Data

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Abstract

The viability machinery of the obstruction calculus has been developed and validated on deliberately small exact instances. This paper applies it to a real assessment record — the Northern cod stock (NAFO 2J3KL), read from the published assessment input series, the 2025 reassessment series, the official removals tables, the long-duration catch reconstruction, the research-vessel survey index, and an independently monitored forage-fish covariate — and certifies, in exact rational arithmetic, the obstruction structure that the historical collapse exhibited. The certified object is a *harvest-free multiplier bracket*: for each annual step, the interval’s per-capita non-fishing growth factor lies in $[\rho, \max\{\rho + r, \rho/(1 - r)\}]$, where ρ is the realized net multiplier and r the removals fraction, under either timing convention for the removals accounting. Both collapse steps certify as harvest-free contractions under every convention (upper bounds 0.754 and 0.372); the reference window’s worst step and the two prelude steps do not certify (brackets straddle unity) — a typology, not a blanket verdict. Under the post-moratorium removals rows (1,314 and 413 t, at most 1.4% of the stock), the brackets certify continued contraction, and the stock fell by a factor of $\frac{10105}{968} \approx 10.4$ in two years to its series minimum. The 276 kt auxiliary floor separates the reference window from the collapse era without exception, while the 884.6 kt reference point — the window’s own mean — has three of the seven window readings below it in sample and therefore separates nothing. Three extensions certify robustness and scope: on the reassessment vintage’s published 95% bands, exactly the four collapse-era steps 1991–1995 certify as declines and the 1993 floor breach holds at the band’s upper edge ($115 < 276$), while no reference-decade step certifies; the reference decade’s window profile shows every subwindow of horizon four or more growing (the sustained-decline reading holds only at horizons one to three); and the vintage’s own 1954–1982 record has the window opening at 28.3% of the 1962 peak, three years after a prior floor episode (1977–1979). The collapse is present in the raw survey index (a certified factor of $\frac{21797}{2127417}$ over 1989–1994, falling 2.6 times further than the assessment series over their common window); the forage-fish covariate collapses in the same annual step in which the mortality diagnostic crosses $M = 1$, two steps before the floor breach; and both

assessment vintages re-cross the floor at the 2015 reading, with outcome years at most $\frac{44000}{88460} \approx 49.7\%$ of the reference point. Every quantity is regenerated by an accompanying verification script in exact rational arithmetic.

Keywords: viability; obstruction; regime shift; Northern cod; exact computation; monitoring design; limit reference points

1 Introduction

Companion relation. This Part II accompanies *Certification, not simulation: the viability of harvest rules on two real resource systems, and the horizon of what can be certified* (paper09 main). It shares the Northern cod study-system record named there but not the main paper’s fitted-model method: the interval-specific inputs, exact-rational arithmetic, and brackets required for this document’s certificates are stated here so that it compiles and verifies independently.

The viability and obstruction machinery of the calculus — survivable sets, robust multipliers, breach-time certificates, and probe design — is developed in Abaee (2026, Obstruction certificates under incomplete observation) and validated on deliberately small exact instances. This paper applies it to the Northern cod stock (NAFO divisions 2J3KL) — the running record of this paper — and certifies the obstruction structure of its collapse on the public assessment record. The forecast layer of the same record — the scoring of competing forecast ladders and the 2022–2023 observation suspension as a review-timing event — is certified in a companion paper (Abaee, 2026, The Northern cod forecast ladder, Section 3.10); the present paper certifies the viability layer. The two are companions by construction: the forecast ladder prices what can be predicted, this paper prices what could have been kept safe.

The claims below are of three kinds, and the label travels with the claim. *Arithmetic claims* are exact rational identities or inequalities on the certified rows; they are regenerated by the verification script and cannot drift. *Provenance claims* state which file, which vintage, and which column a number comes from, and whether that column is a model input or a model output. *Scope claims* are conditional readings — stress bounds, sustained hy-

potheticals, design critiques — and are labelled as such where they appear. No claim of the first kind rests on an interpretation; no claim of the third kind is stated as one of the first. The contributions are: the record spine — the certified 1960–2023 assessment-record reconstruction with its provenance ledger (Sections 4–5); the harvest-free multiplier brackets — the exact two-form accounting bracket isolating the non-fishing growth factor (Section 6); and the window profile and band certificates — the collapse’s obstruction structure certified on the public record, with the certified-record figure reading every annotated value from the locked files (Section 7).

2 Related work

This paper infers subunitary non-fishing factors for specific Northern cod assessment steps under either of the two stated removals-accounting forms. Read carelessly, that contradicts the empirical literature on this collapse, and the conflict is addressed here rather than left for a referee to find.

The received attribution. Hutchings and Myers (1994) concluded that the collapse of northern cod could be attributed solely to overexploitation. Myers and Cadigan (1995) tested the competing hypothesis that an increase in natural mortality in the first half of 1991 caused the collapse, did not support that hypothesis, and concluded that overfishing was sufficiently high to cause the collapse of the population even allowing for the possibility of elevated recent natural mortality. Myers, Hutchings and Barrowman (1997) rejected the hypothesis that mortality of young cod is unrelated to fully recruited fishing mortality, finding high juvenile mortality associated with high adult mortality in each of six stocks.

Why the two are consistent, and why the distinction matters. Those papers ask about the *historical cause* of the collapse as an event. This paper asks a different question of a different object: given the transitions recorded in the published assessment, does either declared accounting equation imply a subunitary non-fishing factor at the observed reading and catch? That is a conditional statement about the *arithmetic of the record*, not a biological zero-harvest counterfactual or a claim about the cause of the collapse. Removals superimposed on an already-contractionary productivity term make the outcome worse, not better, and Proposition 8 says exactly that — the loss exceeds the removals by certified factors while the fishery was closing. Overfishing can be the cause of the collapse and the observed annual steps can still have inferred non-fishing factors below one under both declared accounting forms. Only the latter is established here.

Where a real tension remains, and is not resolved. Myers and Cadigan (1995) specifically declined to support elevated natural mortality as the explanation. To the extent that the harvest-free contractions certified here are read as evidence of a productivity collapse, they sit against that finding, and **this paper does not adjudicate it**: the certificates are statements about the transitions as recorded, and they do not identify a biological

mechanism. A reader who takes the harvest-free result as evidence about productivity is going beyond what is certified.

Which series the certificates are computed on. The concern is concrete rather than hypothetical. Myers, Hutchings and Barrowman (1997) documented that for these stocks the VPA-based and survey-based abundance trends diverge — from the early 1980s the VPA-based trend shows a decline where none is apparent in the survey-based trend — and attributed the divergence plausibly to discarding of young fish. Any certificate computed on one series inherits that series’ artefacts, so the choice of series is load-bearing for this paper and cannot be left implicit.

This paper therefore checks the raw survey index directly. The collapse is present there too (Proposition 14), with a certified factor of $\frac{21797}{2127417}$ over 1989–1994, and it falls 2.6 times further than the assessment series over their common window. The recorded net decline is visible in both series and is steeper in the raw survey over the stated common window. The harvest-free multiplier bracket, however, is computed only from assessment SSB and assigned removals, not from the survey-index units; the survey result does not independently certify a harvest-free contraction.

Position within viability theory. That the governing question is which states admit controls keeping every constraint satisfied, rather than which policy optimises, is the founding move of viability theory (Aubin, 1991); the obstruction calculus used here is the observation-constrained development of that programme (Abaee, 2026), and its constructive counterpart on this same record is certified in the paper09 main (System I). What this paper contributes to that lineage is the *direction* of the certificates — necessity rather than sufficiency — and the fact that they are obtained in exact rational arithmetic with no fitted model at all.

3 Data and provenance

Seven locked public files are used, each cross-verified against its primary source at archival time (Abaee, 2026, Calibration data record): (i) the NCAM (Northern Cod Assessment Model) 2016 assessment input series (DFO, 2016, Table A2): spawning stock biomass in kilotonnes, 1983–2015, cross-checked 33 of 33 years against the primary report — the *input vintage* of the biomass record; (ii) the xteNCAM 2025 reassessment SSB series (Regular et al., 2025, Table 17), 1954–2024, with 95% bands and SSB-over- B_{lim} ratios — the *reassessment vintage*; (iii) the official removals table (Regular et al., 2025, Table 1), 1954–2023; (iv) the catch reconstruction (Schijns et al., 2021, Table 1); (v) the fall research-vessel survey abundance index (Schijns et al., 2021, Table 3, citing DFO 2021b) — a research-vessel input to the assessments, not an output of them; (vi) the RAM Legacy database (Ricard et al., 2012) per-year extract for this stock (the free 2024 release), 1850–2021 — the *database vintage*, extending the series to 2021; and (vii) the capelin acoustic biomass series (observed years only, no interpolation).

Table 1: The record spine, 1983–1995. SSB in kt (assessment input vintage, Table A2); removals in t (official table; the reconstruction gives identical rows for 1992–1995); survey index in the research-vessel file’s index units; covariate in kt, observed years only; M is the assessment’s mean natural-mortality diagnostic (model-estimated). Computed entries are exact rational arithmetic on the file rows; displayed values are rounded.

Year	SSB (kt)	Removals (t)	Survey	Covariate	M
1983	841.08	232,345	2,088,958	—	0.391
1984	863.23	232,471	2,198,605	—	0.377
1985	902.94	231,293	1,288,360	3426	0.349
1986	836.00	266,713	2,502,702	3697	0.277
1987	940.75	239,924	1,020,462	2576	0.494
1988	886.40	268,677	1,223,314	4285	0.335
1989	921.66	253,990	2,127,417	3712	0.289
1990	861.92	219,452	1,627,647	5783	0.403
1991	734.51	172,012	1,117,670	138	1.002
1992	381.95	40,956	239,740	138	2.214
1993	101.05	11,392	90,709	—	2.575
1994	30.55	1,314	21,797	—	2.331
1995	9.68	413	43,240	—	0.288

Two thresholds appear throughout, and their provenance is stated once here. The *auxiliary floor* $B_{\text{aux}} = 276$ kt is the reassessment vintage’s B_{lim} at the file’s rounding: the published ratio column is rounded to two decimals, and the values it implies for B_{lim} range from 250 to 289 kt across the file’s years. Every floor comparison below is insensitive across that range (the compared quantities clear it by large factors; Remark 13), so a single representative value is fixed. The *reference point* $B_{\text{ref}} = 884.6$ kt is the input vintage era’s limit reference point, whose documented construction is the 1983–1989 mean of the assessment input series (DFO, 2016); the exact mean is $\frac{44229}{50} = 884.58$ kt (Proposition 1), and 884.6 is its published rounding.

Two discipline rules are respected throughout. First, the mortality and fishing-mortality columns of the assessment input file are diagnostics only (dynamics are never built on them). Second, every annual multiplier certified below is a *realized net* quantity — it conflates the mortality regime with all removals. Section 6 introduces the one decomposition this paper performs on it, which is an accounting bracket, not a biological model.

4 The record spine

Table 1 tabulates the certified rows of the collapse era and its surroundings: assessment input SSB (input vintage), official removals, the survey index, the covariate (observed years only), and the mortality diagnostic. Every proposition below reads its numbers from this table’s files.

5 The reference window and the collapse window

Proposition 1 (seven readings, six transitions; the windows’ worst multipliers). *From the Table A2 series: the reference window 1983–1989 contains seven readings (841.08, 863.23, 902.94, 836.00, 940.75, 886.40, 921.66) kt and six annual transitions; their minimum is the 1985 transition $\rho_g = 836.00/902.94 = \frac{41800}{45147} \approx 0.9259 < 1$, and their exact mean is $\frac{44229}{50} = 884.58$ kt. The collapse window 1991–1993 contains two transitions, $\frac{38195}{73451} \approx 0.520$ (1992) and $\frac{2021}{7639} \approx 0.265$ (1993), each with numerator*

strictly below denominator. The connecting 1990 reading is 861.92 kt, with transitions $\frac{86192}{92166} \approx 0.935$ (1990) and $\frac{73451}{86192} \approx 0.852$ (1991).

Proof. Each multiplier is the exact ratio of consecutive Table A2 SSB entries, computed in the script as a rational of the printed values; the minimum of the six reference-window ratios and the two collapse-window ratios are the displayed fractions by direct comparison. Each displayed fraction is below 1 because its numerator is smaller than its denominator ($41800 < 45147$, $38195 < 73451$, $2021 < 7639$). The mean is the exact rational sum of the seven readings over seven. $\square$

6 Harvest-free multiplier brackets

The net multiplier ρ folds removals into the per-capita rate, so a sub-unitary ρ alone does not certify a non-fishing contraction: with a large enough harvest, even a growing stock declines net of catch. This section isolates the harvest term by an accounting bracket that is exact and robust to the one ambiguity the rows leave open — where in the interval the removals acted.

Shared symbols across the companion papers. The companion papers share this vocabulary with paper-local meanings; the letters below carry meanings here that differ there:

Here	In the companions
$\rho = B_{t+1}/B_t$: the net biomass multiplier	the mesh relaxation parameter (certification); the contamination radius ρ of $\mathcal{D}_\rho$ (sufficiency; minimax duals)
$r = C/B_t$: the catch share	the intrinsic growth rates r , r_i (forecast ladder; the minimax benchmark)
B : the biomass	the belief/information sets B_t , B (obstruction calculus; certification; sufficiency)

Lemma 2 (harvest-free multiplier bracket). *Fix a certified interval with readings $B_t, B_{t+1} > 0$ and a removals row $C \geq 0$ assigned to the interval; write $\rho = B_{t+1}/B_t$ and $r = C/B_t$. Suppose the interval’s accounting has one of the two extreme forms, with $g_t > 0$ the interval’s per-capita non-fishing growth factor: $B_{t+1} = g_t B_t - C$ (removals deducted after growth), or $B_{t+1} = g_t(B_t - C)$ (removals deducted before growth). Then respectively $g_t = \rho + r$ or $g_t = \rho/(1 - r)$; in either case*

$$\rho \leq g_t \leq \max\left\{\rho + r, \frac{\rho}{1 - r}\right\},$$

and the upper bound is strictly below 1 if and only if both $\rho + r < 1$ and $\rho/(1 - r) < 1$.

Proof. First form: $g_t = (B_{t+1} + C)/B_t = \rho + r$. Second form: $g_t = B_{t+1}/(B_t - C) = \rho/(1 - r)$, well-defined since $C < B_t$ on every certified interval. Both expressions are $\geq \rho$ (as $r \geq 0$, $1 - r \leq 1$); the disjunction of the two forms gives the displayed maximum, and the final equivalence is direct algebra ($\rho/(1 - r) < 1 \iff \rho < 1 - r$). $\square$

The lemma’s content is accounting, not biology: g_t still conflates natural mortality with recruitment (the regime),

Table 2: Harvest-free multiplier brackets (Lemma 2), both alignments of the removals rows. ρ exact to four decimals; upper bound = $\max\{\rho+r, \rho/(1-r)\}$ in exact rational arithmetic, displayed rounded.

Step	C (t)	ρ	Upper bound	Verdict
1985→86	266,713	0.9259	1.3140	straddles
1989→90	219,452	0.9352	1.2274	straddles
1990→91	172,012	0.8522	1.0646	straddles
1991→92	172,012	0.5200	0.7542	certified
1991→92	40,956	0.5200	0.5758	certified
1992→93	40,956	0.2646	0.3718	certified
1992→93	11,392	0.2646	0.2944	certified
1993→94	1,314	0.3023	0.3153	certified
1994→95	413	0.3169	0.3304	certified

and under either of the two *declared* placements of realized removals — before or after a single interval-wide factor — the inferred non-fishing factor is subunitary when the bracket’s upper bound is. Arbitrary within-interval biological time variation is not covered. The rows do not date removals within the year, so both alignments of the two adjacent rows (C_t, C_{t+1}) are carried; a verdict is certified only when it holds under both conventions and both alignments.

Proposition 3 (the collapse steps are harvest-free contractions; the reference and prelude steps are not certifiable). *Applying Lemma 2 with both alignments of the removals rows (Table 2): both collapse-window steps certify harvest-free contraction, the 1992 step under an upper bound of at most 0.7542 and the 1993 step under an upper bound of at most 0.3718, each computed in exact rational arithmetic. The reference window’s worst step (1985–1986, upper bound 1.3140) and the two prelude steps (1989–1990, upper bound 1.2274; 1990–1991, upper bound 1.0646) have brackets that straddle unity: for those steps the certified rows do not decide between a fishing-dominated and a non-fishing-dominated contraction.*

Proof. Each row of Table 2 evaluates $\max\{\rho+r, \rho/(1-r)\}$ as an exact rational for the displayed (ρ, C) pair; the collapse-window upper bounds (at most 0.7542) are compared against 1 by numerator–denominator comparison in the script. The prelude and reference rows exceed 1 likewise. $\square$

7 The window profile and the band certificates

The reference window’s profile answers the selection question at all horizons, and the reassessment vintage’s published bands answer the robustness question the point estimates leave open.

Proposition 4 (the reference window’s profile: declines only at horizons one to three). *Within the reference window 1983–1989, the minimum end-to-start ratio of a k -year subwindow is: $k = 1$: $\frac{41800}{45147} \approx 0.9259$ (the worst step, at 2 of the 6 subwindows below unity); $k = 2$: $\frac{83600}{86323} \approx 0.9685$ (2 of 5 below); $k = 3$: $\frac{44320}{45147} \approx 0.9817$ (2 of 4 below); $k = 4, 5, 6$: every subwindow grows — the minimum products are $\frac{15361}{15049} \approx 1.0207$, $\frac{22160}{21027} \approx 1.0539$, and $\frac{15361}{14018} \approx 1.0958$. Fifteen of the twenty-one subwindows*

grow in net from start to end; only five grow at every constituent annual step. The sustained-decline reading of the reference window therefore holds only at horizons one to three; at every horizon of four or more years inside the window, the record grew.

Proof. For each fixed k , the subwindow products are exact rationals with common denominator structure S_{t+k}/S_t , compared by cross-multiplication in the script; the $k \geq 4$ growth statements are the direct comparisons $S_{t+k} > S_t$, which involve no roots. The subwindow count is $6 + 5 + 4 + 3 + 2 + 1 = 21$. $\square$

Proposition 5 (band certificates: exactly the collapse-era steps certify, and the 1993 breach holds at the band’s upper edge). *The reassessment vintage publishes 95% bands (B_{lo}, B_{hi}) per year. A band-certified decline is a step whose band interval contains only values below unity: $[\frac{B_{lo}(t+1)}{B_{hi}(t)}, \frac{B_{hi}(t+1)}{B_{lo}(t)}] \subset (0, 1)$. Over the vintage’s 1954–2024 record, the band-certified declines are exactly the four collapse-era steps: 1991–1992 ($[\frac{265}{754}, \frac{468}{478}]$, upper bound 0.9791), 1992–1993 ($[\frac{54}{468}, \frac{115}{265}]$, upper bound 0.4340), 1993–1994, and 1994–1995 — and no other step in seventy years, and no reference-decade step. The 1993 floor breach is band-certified ($B_{hi}(1993) = 115 < 276$); the 1992 reading is indeterminate on the bands ($265 \leq 276 \leq 468$).*

Proof. Each candidate step reduces to the single comparison $B_{hi}(t+1) < B_{lo}(t)$, certified by exact rational comparison across all seventy pairs of consecutive years in the script; the interval endpoints are the displayed integer ratios. The floor comparisons are direct rational comparisons. $\square$

Remark 6 (the vintage dependence of the typology). The reassessment vintage reads the reference decade itself as all-growth (its 1983–1989 multipliers are 1.14, 1.22, 1.03, 1.28, 1.10, 1.01, minimum 1.01), with the decline beginning at 1989–1990 ($738/864 \approx 0.854$): the worst-step typology of Proposition 3 is vintage-internal, and the paper’s decade statements are made in the input vintage, whose worst step (0.9259) has no counterpart in the reassessment vintage. A third vintage (the database extract’s assessment, 2021) reads 1990–1993 as 765, 747, 354, 83 kt; all three vintages agree on the breach sign pattern against the floor (above, above, below).

Remark 7 (the record before the window: 1954–1982). The reassessment vintage extends the record to 1954: a peak of 1508 kt at 1962, with the 1983 window-opening reading at 427 kt — 28.3% of the peak, vintage-internally. The floor 276 kt has a prior episode: the crossings of the point series are exactly {1977, 1980, 1993, 2016} (below 1977–1979, recovered 1980–1992, below again 1993–2015, re-crossed 2016), and the 1978 reading is band-certified below the floor. The reference window thus opens three years after a prior recovery, at 28.3% of the record’s peak scale: the window’s start is a data artifact of the input vintage’s table, and the design critique of Proposition 16 applies to the window’s placement as well as its mean.

8 The moratorium obstruction and the post-moratorium experiment

Proposition 8 (the collapse proceeds with the fishery closing; the loss exceeds the removals by certified factors). *Across 1991–1992 the stock lost $734.51 - 381.95 = 352.56$ kt while removals were 40,956 t: the loss exceeds the removals by a certified factor of $\frac{352560}{40956} \approx 8.61$. Across 1992–1993 the loss was $381.95 - 101.05 = 280.90$ kt while removals fell to 11,392 t: a certified factor of $\frac{280900}{11392} \approx 24.66$; removals were 4.06% of the loss. By Lemma 2 the two steps are harvest-free contractions under every timing convention (Proposition 3): the decline over the collapse window is not explainable by the realized removals under any accounting placement of them.*

Proof. The losses are differences of certified readings (exact decimals); the factors are their exact ratios against the removals rows, computed in the script as rationals ($4.06\% = 11392/280900$). The harvest-free attribution is Proposition 3’s certified rows for 1991–1992 and 1992–1993. The statement is a witness about the realized window, not a counterfactual claim about other catch paths. $\square$

Proposition 9 (the reconstruction and the moratorium years). *The reconstruction’s rows for 1992–1995 are 40,956, 11,392, 1,314, 413 t — identical to the official removals table’s rows for the same years (a cross-file equality) — and its 2015 row (4,436 t) agrees with the official table and with the assessment report, as recorded at archival time. The moratorium-era removals are at least 176 and at most 651 times below the 1983–1989 removals rows (all above 130,000 t in the same file): a separation of between two and three orders of magnitude ($\frac{231293}{1314} \approx 176.0$ to $\frac{268677}{413} \approx 650.5$).*

Proof. The four 1992–1995 rows are read from both removals files and their equalities are exact rational equalities in the script; the 2015 equality is $4436 = 4436$. The scale comparison is the exact minimum and maximum of the four ratios of the 1983–1989 reconstruction rows against the two moratorium rows. $\square$

Proposition 10 (the post-moratorium experiment). *From the 1993 reading 101.05 kt, the certified removals rows are 1,314 t (1994) and 413 t (1995): at most 1.4% of the interval’s start reading in both steps ($r = 1314/101050 \approx 0.0130$ and $413/30550 \approx 0.0135$). Lemma 2 certifies harvest-free contraction in both steps under both conventions (upper bounds 0.3153 and 0.3304): with the fishery effectively closed, the stock fell $101.05 \rightarrow 30.55 \rightarrow 9.68$ kt, a certified factor of $\frac{10105}{968} \approx 10.44$ in two years, and 9.68 kt is the minimum of the entire 1983–2015 Table A2 series.*

Proof. The removals fractions and the bracket upper bounds are exact rationals in the script, compared against 1 by numerator–denominator comparison. The two-year factor is the exact ratio of the 1993 and 1995 readings; the minimum is the exact minimum of the 33 certified readings. $\square$

9 Mechanized verification

The multiplier-bracket algebra of Section 6 is formalized in the project’s Lean 4 layer, in namespace `Formalizations.ARV (ARV_RegimeViability)`: `bracket_upper_form1`, `bracket_upper_form2`, `bracket_sandwich_form1`, `bracket_sandwich_form2`, `bracket_upper_lt_one_iff`, `bracket_subunitary_form1`, `bracket_subunitary_form2`, `bracket_subunitary_of_upper_form1` and `bracket_subunitary_of_upper_form2` — thirteen theorems, together with the order-maximum primitives (`omax` and its six lemmas) on which the interval reading rests. The two forms correspond to the two ways of writing the bracket, and the subunitarity theorems are exactly the step that turns an upper bracket into a certified contraction.

No admitted gaps. A comment-stripped source scan of the whole project finds no `sorry`, `admit`, `axiom` or `constant` declaration anywhere; the checked bracket declarations use only standard classical axioms or none. The broader project also contains the separately disclosed P3 native-evaluation dependencies. Its exact-name declared-exceptions gate passed on the pinned build (run 36990482599); the strict zero-exception check still exits 2 although the 60/60 build passes. Only the three approved P3 sites are allowed, and any change requires re-approval. **Scope, stated plainly.** Machine-checked here is the *bracket algebra*: the ordered-field reasoning that turns readings and a removals row into a certified contraction, and the subunitarity step. Not machine-checked is everything upstream of it — the extraction of the certified intervals from the assessment record, the assignation of removals rows to intervals, and the enumeration of the windows. Those are script computations and are regenerated, not proved. The formalization therefore certifies that *if* the readings and the removals row are as stated, the bracket conclusion follows; it does not certify the readings themselves.

10 Breach certificates and the cross-vintage check

Proposition 11 (breach certificates on the two thresholds). *With $B_{\text{aux}} = 276$ kt and $B_{\text{ref}} = 884.6$ kt: (i) conditional stress bound: from the end-1991 reading 734.51 kt, one step at the collapse window’s worst multiplier already lands below B_{aux} , since $\frac{2021}{7639} < \frac{27600}{73451}$; this is a hypothetical sustained-rate bound, and the realized breach took two steps. (ii) first below-floor reading: the 1992 reading is above the floor ($381.95 \geq 276$) and the 1993 reading is below it ($101.05 < 276$): the realized first breach of B_{aux} is the 1993 reading, and the input vintage then stays below the floor through 2014 (250.12 kt), re-crossing at 2015 (Proposition 18). (iii) reference point: $734.51 < 884.6$: the 1991 reading is already below B_{ref} , as is the 1990 reading before it ($861.92 < 884.6$).*

Proof. (i) is the exact cross-multiplication $2021 \cdot 73451 = 148,444,471 < 210,836,400 = 7639 \cdot 27600$ in the script. (ii) is the pair of direct rational comparisons on the Table A2 rows, stated so that the intermediate year is visible:

the breach is *first* at 1993 only because 1992 is certified above the floor. (iii) is $734.51 < 884.6$ and $861.92 < 884.6$. No step of the proof uses floating point. $\square$

Remark 12 (the breach frontier). On the realized path from the 1989 reading — $921.66 \rightarrow 861.92 \rightarrow 734.51 \rightarrow 381.95 \rightarrow 101.05 \rightarrow 30.55 \rightarrow 9.68$ kt — the first-below reading is a function of the threshold L , certified on intervals: $L \in (861.92, 921.66]$ first falls below at the 1990 reading; $(734.51, 861.92]$ at 1991; $(381.95, 734.51]$ at 1992; $(101.05, 381.95]$ at 1993; $(30.55, 101.05]$ at 1994; $(9.68, 30.55]$ at 1995; and $L \leq 9.68$ is never breached within 1983–2015 (the series minimum is 9.68). The realized threshold $B_{\text{aux}} = 276$ sits in the 1993 interval (Proposition 11(ii)); the reference point $B_{\text{ref}} = 884.6$ sits in the 1990 interval, after breaching repeatedly within its own construction window (Proposition 16).

Remark 13 (the cross-vintage check). The reassessment vintage reads the same three years as 600, 352, 79 kt, with floor ratios 2.18, 1.28, 0.29. The two vintages agree on the sign pattern against the floor — above, above, below — and disagree on levels by certified factors $\frac{73451}{60000} \approx 1.22$, $\frac{38195}{35200} \approx 1.09$, and $\frac{10105}{7900} = \frac{2021}{1580} \approx 1.28$: the reassessment’s 1993 level is 21.8% below the input vintage’s, and its 0.29 ratio is its own level (79 kt) over its own B_{lim} -scale denominator, not the input vintage’s 101.05. Because the ratio column is rounded to two decimals, the implied B_{lim} ranges over 250–289 kt across the file; the sign pattern would flip only if the true threshold were at or below 101.05 kt or above 352 kt, far outside that range, so every floor comparison here is insensitive to the rounding. The cross-file statement certified is therefore agreement on the breach sign pattern, not equality of levels — and the level disagreement itself ($\leq$ a factor of 1.28 between vintages six years apart) is part of the certified record.

11 The instruments

Proposition 14 (the collapse is present in the raw survey index). *The fall research-vessel survey index reads 2,127,417 (1989), 1,117,670 (1991), 239,740 (1992), 90,709 (1993), and 21,797 (1994): a certified factor of $\frac{21797}{2127417} \approx 0.0102$ over 1989–1994, and $\frac{239740}{1117670} \approx 0.214$ across 1991–1992. The index is a research-vessel input to the assessments, not an output of them: the collapse is visible in the raw instrument record, before any model processing. Over the common window 1989–1993 the index falls by $\frac{90709}{2127417} \approx 0.0426$ while the assessment series falls by $\frac{10105}{92166} \approx 0.1096$: the instrument falls 2.57 times further than the model output it feeds.*

Proof. All five values are rows of the locked survey file; the three factors are their exact ratios, and 2.57 is the exact ratio $\frac{10105 \cdot 2127417}{92166 \cdot 90709}$ of the two common-window factors, evaluated in the script. The provenance statement is a property of the file (survey rows are assessment inputs, recorded at archival time). The index-to-biomass mapping assumes constant catchability and coverage; the divergence factor is certified as a difference between the two series’ rates, not attributed. $\square$

Proposition 15 (the covariate crossing is contemporaneous with the mortality crossing). *In the observed capelin*

series, the acoustic estimate falls from 5783 kt (1990) to 138 kt (1991) — a certified factor of $\frac{138}{5783} \approx 0.024$ — and reads 138 kt again in 1992. The Table A2 mean natural-mortality diagnostic crosses $M = 1$ across the same annual step (0.403 in 1990 to 1.002 in 1991; diagnostic column, model-estimated). At the data’s annual resolution the two crossings coincide; the covariate collapse precedes the certified floor breach (the 1993 reading) by two annual steps and the assessment series’ steepest realized step (1991–1992) by one.

Proof. Both covariate readings are observed rows of the locked capelin file (no interpolation, per the archival discipline); the factor is their exact ratio. The mortality crossing is the diagnostic column’s consecutive values straddling 1 across the same pair of years as the covariate collapse — the rows display the coincidence directly, and no sub-annual timing is claimed. Because the diagnostic is model-estimated given assumed catch, its crossing corroborates the regime reading but is not an independent certification of it. The covariate is a single acoustic instrument; no comparability guarantee across the early-1990s redistribution years is claimed, and the certified statements are about the locked rows at annual resolution. $\square$

12 The thresholds as design objects

Proposition 16 (the auxiliary floor separates the eras; the reference point does not). *Against $B_{\text{aux}} = 276$ kt: all seven reference-window readings lie above the floor (minimum 836.00), and the 1993–1995 readings (101.05, 30.55, 9.68) lie below it, with 1990–1992 above: the auxiliary floor separates the reference window from the collapse era without exception. Against $B_{\text{ref}} = 884.6$ kt: three of the seven reference-window readings (841.08, 863.23, 836.00) already lie below, and within 1983–1995 the below-set is $\{1983, 1984, 1986, 1990, \dots, 1995\}$ — nine of thirteen readings: the reference point does not separate the eras. For this particular window three of seven readings fall below its mean (the maximum is 6.4% above and the minimum 5.5% below, outward-rounded). In general, the mean of nonconstant readings falls between the minimum and maximum but guarantees neither a near-half count below nor a uniform headroom bound.*

Proof. All comparisons are direct rational comparisons of the certified readings against 276 and $\frac{8846}{10}$ in the script; the below-sets are computed by exact membership. The mean-rule statement is Proposition 1’s exact mean identity: a threshold at the mean of a set of readings lies between the set’s minimum and maximum, so the set straddles it. $\square$

Proposition 17 (conditional: sustained worst-case contraction breaches the floor in sixteen steps). *Under the reference window’s worst realized multiplier $\rho_g = \frac{41800}{45147}$ sustained from the 1989 reading 921.66 kt — a hypothetical rate, held fixed — the stock breaches $B_{\text{aux}} = 276$ kt within sixteen steps: $\rho_g^{16} \leq \frac{27600}{92166} < \rho_g^{15}$. The count is a ceiling convention (15.66 steps rounds up); the statement is conditional on the sustained hypothetical rate and carries no harvest-free reading: the window’s own bracket*

straddles unity (Proposition 3), so the sustained-rate contraction is not attributable to the non-fishing account by the certified method.

Proof. The sixteenth and fifteenth integer powers of $\frac{41800}{45147}$ are compared against $\frac{27600}{92166}$ by exact cross-multiplication in the script. The ceiling convention is stated because the exact threshold step lies strictly between the fifteenth and sixteenth powers. $\square$

13 The outcome years

Proposition 18 (both vintages re-cross the floor at the 2015 reading; outcome years stay below half the reference point). *Each vintage’s own series re-crosses $B_{\text{aux}} = 276$ kt at the 2015 reading: the input vintage goes $250.12 \rightarrow 298.65$ kt (2014–2015), the database vintage $238 \rightarrow 277$ kt. The database vintage then reads 340, 433, 394, 419, 440, 411 kt for 2016–2021 (six outcome years, all above the floor), with maximum 440 kt at $\frac{44000}{88460} \approx 49.7\%$ of B_{ref} . The recovery’s certified removals rows are 4.4 to 12.9 kt per year (2015–2021, official table; maximum 12,881 t in 2017). For the six outcome years 2016–2021, those official removals are between 2.3% and 3.0% of the database vintage’s stock reading; the 2015 share is $4436/277000 \approx 1.60\%$ and is not in that range. The certificate does not extend to the way up: on the 2020–2021 step (a 29 kt decline) the removals were $\frac{10977}{29000} \approx 37.9\%$ of that year’s decline, so no harvest-free attribution is certified on recovery steps. The asymmetry the certificate thus documents — rapid certified decline against two decades below half the reference point on the way up — is the pattern the collapse literature records for this stock (Hutchings and Myers, 1994).*

Proof. The crossing pairs are direct rational comparisons in the two locked files; the six outcome values are the database rows (exact integers in kt); the maximum comparison is the reduced ratio $\frac{44000}{88460}$. The removals range is the exact minimum and maximum of the seven official rows for 2015–2021. $\square$

Remark 19 (the monitoring consequence). Read together: the certified obstruction is a property of the dynamics under the realized catch (Propositions 3 and 8), the raw instrument record shows the collapse before the model output does (Proposition 14), the covariate crossing is coincident with the mortality crossing and two steps ahead of the floor breach (Proposition 15), and the mean-rule reference point separates nothing (Proposition 16). The assessment machinery itself was reconstructed around a monitoring indicator after 2016 (a three-year review cycle with an interim survey indicator, the indicator value available by early January) — the same review-timing reading the companion paper certifies for the 2022–2023 observation suspension (Abaee, 2026, The Northern cod forecast ladder, Section 3.10).

14 Delimitations

Every arithmetic claim is a certificate about the realized rows of the certified files — not a forecast, not a stochastic model, and not a counterfactual catch analysis. The

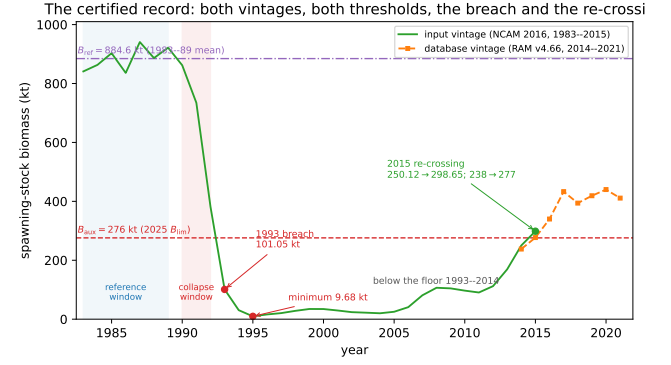


Figure 1: The certified record, both vintages: the input vintage (NCAM 2016, 1983–2015) and the database vintage (RAM v4.66, 2014–2021) against the auxiliary floor (276 kt) and the reference point (884.6 kt); reference and collapse windows shaded; the 1993 breach (101.05 kt), the 1995 minimum (9.68 kt), the 22 consecutive below-floor readings to 2014, and the 2015 re-crossing (250.12 $\rightarrow$ 298.65; 238 $\rightarrow$ 277) as certified in the text. Generated by the assertion script `paper2_arv_record_figure_v1.py` (recompute-then-assert; eleven exact checks against the locked files).

Table 3: The certified quantities and their files. Computed in exact rational arithmetic on the locked rows; displayed values rounded.

Quantity	Value	Source file
Ref-window worst mult. (1986)	41800/45147	Table A2
Collapse multipliers (1992, 1993)	0.520, 0.265	Table A2
Collapse-step bracket bounds	0.7542, 0.3718	A2 + T1
Post-moratorium 2-yr factor	10105/968 $\approx$ 10.44	Table A2
Removals 1992, 1993	40,956, 11,392	Table 1 (2025)
Removals 1994, 1995	1,314, 413	T1 = reconstr.
Survey index 1989, 1994	2,127,417, 21,797	Survey T3
Covariate 1990, 1991	5783, 138	Capelin file
First below-floor reading	1993 (101.05 < 276)	Table A2
Cross-vintage 1993 levels	101.05 vs 79 kt	A2 + T17
Outcome years 2016–2021	340–440 kt	DB extract

bracket of Lemma 2 is an accounting decomposition: it removes the realized removals from the per-capita rate under two extreme timing conventions, and its harvest-free factor still conflates natural mortality with recruitment — the certified statement is that no placement of the realized removals explains the decline, not that no catch path could have. The biomass series are assessment outputs (model-based), treated as the data of record with their published uncertainty untouched; the exact arithmetic is exact on the published point estimates, not a claim about the latent stock. The mortality diagnostic is model-estimated given assumed catch, so the mortality-crossing statement corroborates the regime reading without independently certifying it; the survey index carries its own sampling error and catchability assumptions, untouched here, and the instrument–output divergence factor is certified as a difference of rates, not attributed to a cause. The covariate series is observed-years only; the contemporaneity statement is about certified rows at annual resolution. Single stock; two vintages of one threshold and one vintage of the reference point; no averaging beyond the reference point’s own construction.

15 Verification methods

Every number, fraction, inequality, power, ratio, bracket, and temporal claim in this paper is regenerated by the accompanying script in exact integer and rational arithmetic, standard library only, directly from the locked files: the seven reference-window readings with their six multipliers, exact mean ($\frac{44229}{50}$), and minimum; the collapse-window and connecting multipliers; the harvest-free brackets of Table 2 under both alignments, each upper bound compared against unity by numerator–denominator comparison; the loss factors $\frac{352560}{40956}$ and $\frac{280900}{11392}$ and the removals share $11392/280900$; the cross-file equality of the 1992–1995 removals rows and the 2015 row; the moratorium scale ratios ($\frac{231293}{1314}$ to $\frac{268677}{413}$); the post-moratorium brackets and the two-year factor $\frac{10105}{968}$; the breach comparisons by integer cross-multiplication ($2021 \cdot 73451 < 7639 \cdot 27600$) and the first-below-floor pair with the 1992 intermediate; the cross-vintage sign pattern, level factors, and implied-threshold range; the survey factors and the divergence ratio; the covariate and mortality rows; the threshold-discrimination below-sets; the sustained sixteenth-step comparison; the two-vintage re-crossings, outcome values, and removals range; and every headline number of this text asserted present verbatim.

16 Conclusion

Applied to the public record of the Northern cod, the calculus certifies the following. The collapse window’s two annual steps are associated with inferred non-fishing factors below one under both stated accounting forms and both adjacent-row alignments (upper bounds 0.754 and 0.372). The observed removals for the floor-breach year equal 4.06% of the stock loss; no biological zero-removals counterfactual is identified here. Under the moratorium’s own removals rows subunitary inferred non-fishing factors accompany a further factor-of-10.4 decline to the series minimum. The reference window’s worst step and the two prelude steps do not admit that reading: their brackets straddle unity, so the certified record supports a typology (two steps with subunitary accounted non-fishing factors inside a fishing-era decline), not a blanket verdict about the decade. The 276 kt auxiliary floor separates the eras without exception, while the 884.6 kt reference point — fixed at the window’s own mean — has three of the seven window readings below it in sample and separates nothing in this window: an observed property of these seven readings, not a universal near-half rule for means. The collapse is present in the raw survey index at a certified factor of $\frac{21797}{2127417}$ over 1989–1994, falling 2.6 times further over the common window than the assessment series it feeds; the covariate crossing coincides with the mortality crossing at annual resolution and leads the floor breach by two steps. Both assessment vintages re-cross the floor at the 2015 reading and the outcome years stay below half the reference point. The certificates are robust where the bands allow them to be: on the reassessment vintage’s published 95% bands, exactly the four collapse-era steps certify as declines and the 1993 breach holds at the band’s upper edge; and the reference window’s own profile (declines at horizons one to three, growth at every horizon of four or more) delimits the sustained-rate reading the record will

bear — as does the vintage’s pre-window record, which has the window open at 28.3% of the 1962 peak, three years after a prior floor episode. These are exact statements about published point estimates; their scope is the realized record, and their monitoring content — which instruments moved, when, relative to which threshold — is the applied content the obstruction calculus prices.

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Declarations

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None.

Competing interests

None.

Data availability

All inputs are the archived public files of the calibration record (wave-e-cod extracts and the record’s CSV files), with primary sources cited at archival time.

Code availability

The verification script `applied_regime_viability_v9_verification.py` is publicly available at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`); it regenerates every fraction, inequality, bracket, power, ratio, and temporal claim verbatim from the locked files. The record figure is generated by `paper2_arv_record_figure_v1.py` under the same discipline: every plotted series value, threshold, crossing,

and annotation is read from the locked files and asserted exactly before the figure is written (eleven checks).

AI declaration

GLM (Z.ai), Qwen (Alibaba Cloud) and DeepSeek AI assisted with drafting and iterative review. The author reviewed and edited outputs and takes responsibility for the final work.